Bioinformatics for Beginners

Introduction to Bioinformatics: An Overview

Tugce Bilgin Sonay, ZHAW

Meet up

1 no bioinformatics experience

2 have used an online tool or two

Question:
Which analyses/approaches are used in bioinformatics? Anything you've done or heard before... Be specific. What is the point of bioinformarmatics?

3 have worked with tools in the terminal

4 develop tools

Meet up

1 interested in DNA and mutations

2 interested in single cell + spatial transcriptomics

Question:
What would you like to get from this course?

3 interested in tools

4 interested in clinical outcomes

Meet us

Research Centre for Bioinformatics

We develop practical solutions at the interface of biology, medicine and computational sciences.

The Research Centre for Bioinformatics focuses on the theoretical and computational aspects of modelling the molecular biology processes, genome evolution and adaptive change, as well as biomedical data representation and integration. The goal is to bring basic research and new bioinformatics methods to real-world applications, ranging, for example, from biotechnology and forensics to biomedical research and environmental applications. The research area is represented by the several research groups, each focusing on certain methods or application domains.

Meet us

Computational Genomics Lab:

Maria Anisimova

Olesia Kondrateva

Feifei Xia

Inbar Leaf

Tugce Bilgin Sonay

Research Centre for Bioinformatics at ZHAW

Computational Genomics

The research group develops computational methods for comparative and evolutionary genomics, including modelling of stochastic processes in molecular evolution. Many research projects focus on the analysis of protein-coding genes and gene families, selection, adaptation, phylodynamics and evolution, including host-pathogen interactions; applications in medical genomics, epidemiology, metagenomics and forensics. Our research includes studies of genomic repeat sequences and indel evolution with applications in cancer genomics and biotechnology, as well as studies of dynamics and evolution of viruses and other pathogens.

Group leader: Prof. Dr. Maria Anisimova

https://github.com/acg-team/Bioinfo4B

Research Centre for Bioinformatics at ZHAW

Biomedical String Analysis

The research group is specialized in the analysis of strings (i.e. finite sequences of symbols). The research projects and applications focus on genomic data and biomedical natural language. The group develops new computational science methods and applies existing methods. This includes: mathematical modeling, computational statistics, algorithm design, discrete mathematics, machine and deep learning, natural language processing, semantic web technologies.

Group leader: Dr. Manuel Gil | Learn more about the research group Biomedical String Analysis

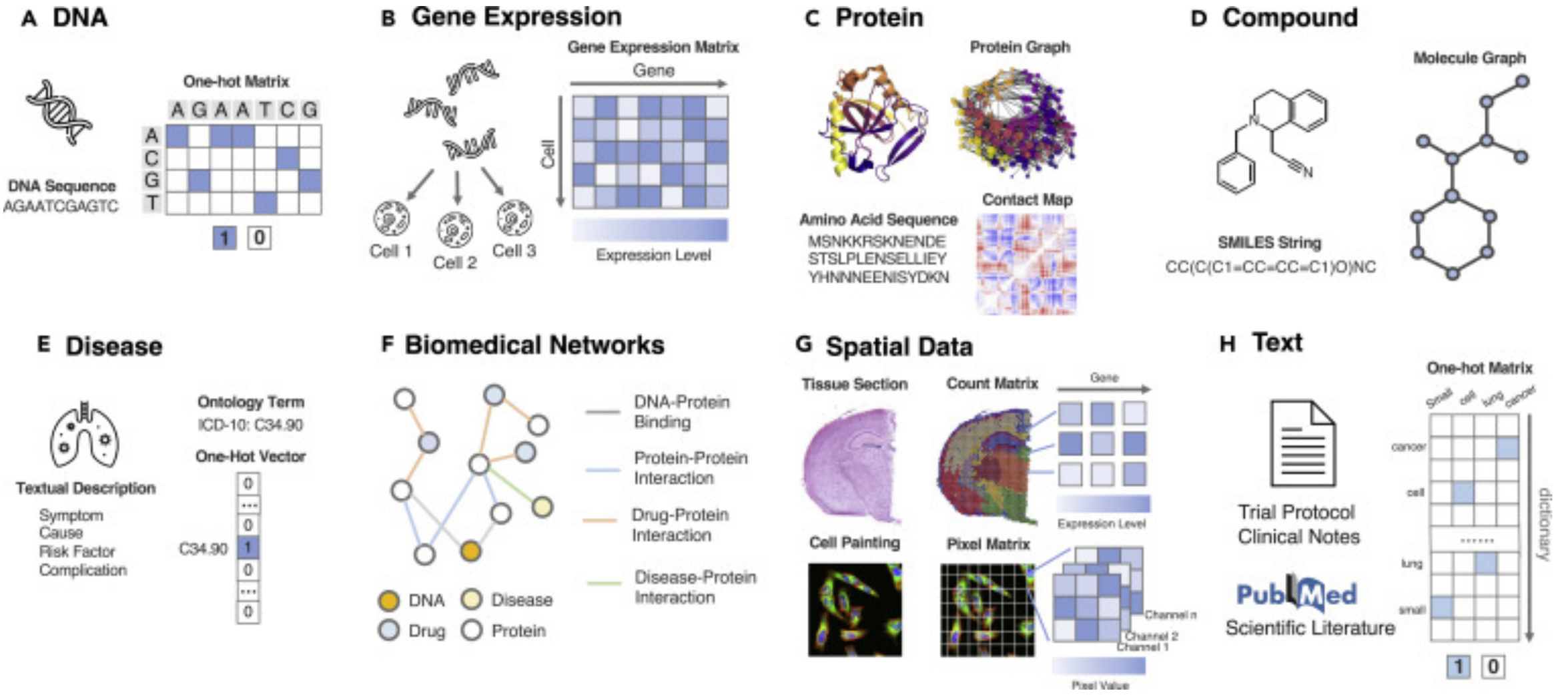
Applied Mathematical Biology

The research group develops and applies mathematical models and methods to address open research questions in biology. Many methods use standard calculus, differential equations, machine learning and dynamical systems theory to describe and predict biological phenomena, such as for example, the relationship between codon bias and gene expression via the concept of translational efficiency, applied to codon optimization problems. Further interests lie in the exploration of cancer-immune system interactions and their predictive power for cancer immunotherapies as well as the population genetics of the early infection-phase of partially-recombining viruses.

Group leader: Dr. Victor Garcia

Bioinformatics

Zürcher Hochschule für Angewandte Wissenschaften
zh aw Life Sciences und Facility Management
ICLS Institut für Computational Life Sciences



Bioinformatics Resources

Protein Databases
- Protein-protein interaction: https://string-db.org/
- Protein databases: https://www.uniprot.org/ https://www.rcsb.org/

Bioinformatics Forums
- Best friend of a bioinformatician: https://www.biostars.org
- Another best friend: https://stackoverflow.com

Notebooks
- Jupyter Notebook, bioinformatician's diary: https://jupyter.org

Downstream Genomic
- Bedtools : https://bedtools.readthedocs.io/en/latest/
- USCS Table Browser: https://genome.ucsc.edu/cgi-bin/hgTables

Omics Databases
- Biomart Ensembl: https://useast.ensembl.org/index.html
- ncbi entrez: https://www.ncbi.nlm.nih.gov/

Cancer Databases
- Pan Cancer, TCGA: https://cancergenome.nih.gov

Gene Databases
- Gene Cards: https://www.genecards.org
- DAVID for Gene Ontology terms: https://david.ncifcrf.gov/summary.jsp

Genome Databases

1000 genomes new name : https://www.internationalgenome.org

Coding
- R Circos - https://cran.r-project.org/web/packages/BioCircos/vignettes/BioC
- R packages: https://www.bioconductor.org
- python packages: Biopython